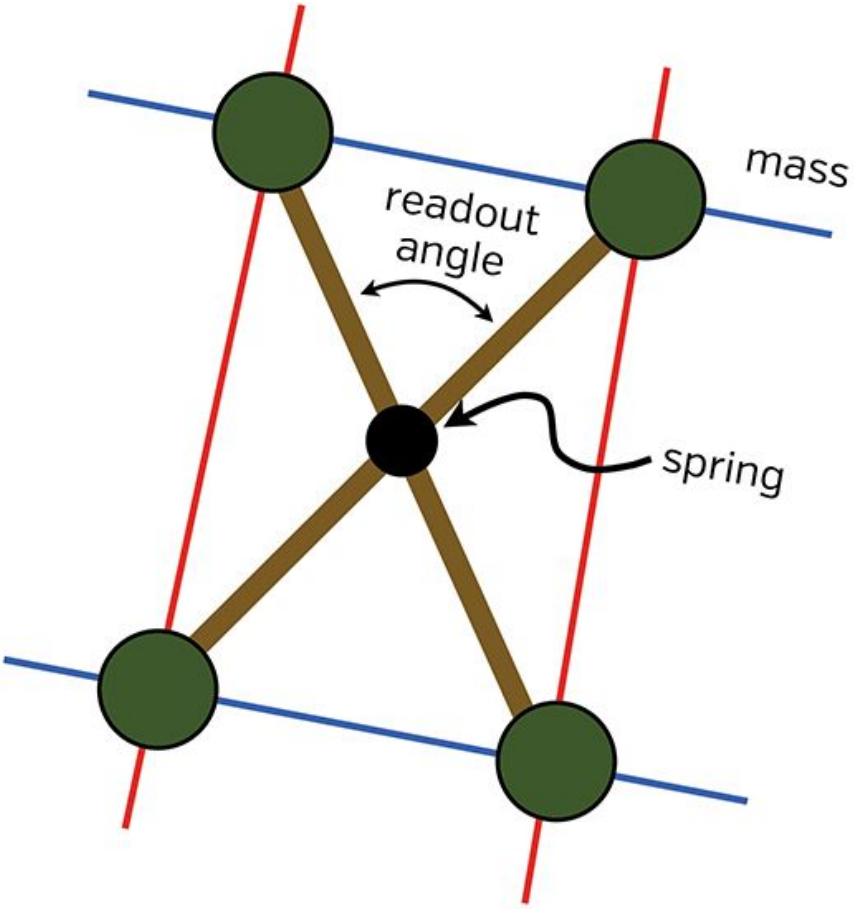


Fig. 24.7. A simple version of a gravity gradiometer, designed and built by Robert Forward at Hughes Research Laboratories in 1970.

If this gradiometer is flown rightward through the tidal pattern of Figure 24.6, its readout angle opens up above the oil-bearing region, and then closes down over the oil-free region. Gradiometers like this, but more sophisticated, are used by geologists to search for oil and also for mineral deposits.

NASA has flown a more sophisticated gradiometer called GRACE[40] (Figure 24.8) to map tidal fields everywhere above the Earth, and watch slow changes of tidal gravity produced, for example, by the melting of ice sheets.